Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

_____ is formed when potassium iodide is heated with conc. H_2SO_4

(A) HI

(B) I_2

(C) HIO_3

(D) KIO_3

Q.2

Sodium borate on reaction with conc. $H_2\ SO_4$ and C_2H_5OH gives a compound (A) which burns with a green edged flame. the compound (A) is:

(A) H_2BuO_7

(B) $(C_2H_5)_2\ B_4O_7$

(C) H_3BO_3

(D) $(C_2H_5)_3\,BO_3$

Q.3

A solution of a salt in dilute sulphuric acid imparts deep blue colour with starch iodine solution it confirms the presence of which of the following

(A) NO_2^-

(B) I^-

(C) NO_3^-

(D) CH_3COO^-

Q.4

$[x] + H_2SO_4 \longrightarrow [y]$ a colourless gas with irritating smell; $[y] + k_2Cr_2O_7 + H_2SO_4 \rightarrow$ green solution. $[x]$ and $[y]$ is:

(A) SO_3^{2-} and SO_2

(B) Cl^- and HCl

(C) S^{2-} and $H_2\ S$

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(D) CO_3^{2-} and CO_2

Q.5

When KBr is treated with conc. H_2SO_4 a reddish-brown gas is evolved. The evolved gas is

(A) Bromine

(B) Mixture of bromine and HBr

(C) HBr

(D) NO_2

Q.6

An aqueous solution of a salt X turns blood red on treatment with CNS^- and blue on treatment with $K_4[Fe(CN)_6]$. X also gives a positive chromyl chloride test. The salt X is :

(A) $CuCl_2$

(B) $FeCl_3$

(C) $Cu(NO_3)_2$

(D) $Fe(NO_3)_3$

Q.7

The cation that will not be precipitated by H_2S in the presence of dil. HCl is :

(A) Pb^{2+}

(B) Cu^{2+}

(C) Co^{2+}

(D) As^{3+}

Q.8

An aqueous solution of a substance gives a white precipitate on treatment with dil. HCl which dissolves on heating. When H_2S is passed through the hot acidic solution, a black precipitate is obtained. The substance is a

(A) Hg_2^{2+} salt

(B) Cu^{2+} salt

(C) Ag^+ salt

(D) Pb^{2+} salt

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Q.9

A metal chloride dissolves appreciably in cold water. When placed on a platinum wire in Bunsen flame no distinctive colour is noticed, the cation would be

(A) Mg^{2+}

(B) Ba^{2+}

(C) Pb^{2+}

(D) Ca^{2+}

Q.10

In the separation of Cu^{2+} and Cd^{2+} in 2 nd group qualitative analysis of cations, tetrammine copper (II) sulphate and tetrammine cadmium (II) sulphate react with KCN to form the corresponding cyano complexes. Which one of the following pairs of the complexes and their relative stability enable the separation of Cu^{2+} and Cd^{2+}?

(A) $K_3\,[Cu(CN)_4]$ more stable and $K_2\,[Cd(CN)_4]$ less stable

(B) $K_2\,[Cu(CN)_4]$ less stable and $K_2\,[Cd(CN)_4]$ more stable

(C) $K_2\,[Cu(CN)_4]$ more stable and $K_2\,[Cd(CN)_4]$ less stable

(D) $K_3\,[Cu(CN)_4]$ less stable and $K_2\,[Cd(CN)_4]$ more stable

Q.11

Sodium carbonate cannot be used in place of $(NH_4)_2\,CO_3$ for the identification of Ca^{2+}, Ba^{2+} and Sr^{2+} ions (in group V) during mixture analysis because :

(A) Mg^{2+} ions will also be precipitated.

(B) Concentration of $CO_3{}^{2-}$ ions is very low.

(C) Sodium ions will react with acid radicals.

(D) Na^{+} ions will interfere with the detection of $Ca^{2+}, Ba^{2+}, Sr^{2+}$ ions.

Q.12

A solution containing $As^{3+}, Cd^{2+}, Ni^{2+}$ and Zn^{2+} is made alkaline with dilute NH_4OH and treated with $H_2\,S$. The precipitate obtained will consist of

(A) $As_2\,S_3$ and CdS

(B) CdS, NiS and ZnS

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(C) NiS and ZnS

(D) Sulphide of all ions

Q.13

Which of the following statements is incorrect?

(A) Fe^{2+} ion also gives blood red colour with SCN^- ion.

(B) Fe^{3+} ion also gives blood red colour with SCN^- ion.

(C) On passing H_2S into Na_2ZnO_2 solution a white ppt of ZnS is formed.

(D) Cupric ion reacts with excess of ammonia solution to give deep blue colour of $[Cu(NH_3)_4]^{2+}$ ion.

Q.14

An aqueous solution of colourless metal sulphate M gives a white precipitate with NH_4OH. This was soluble in excess of NH_4OH. On passing H_2S through this solution a white ppt. is formed.
The metal M in the salt is

(A) Ca

(B) Ba

(C) Al

(D) Zn

Q.15

For preparing $250\ mL$ of $N/20$ solution of Mohr's salt, the amount of Mohr's salt needed is

(A) $9.8\ g$

(B) $4.9\ g$

(C) $19.6\ g$

(D) $3.2\ g$

Q.16

Three separate samples of a solution of a single salt gave these results. One formed a white precipitate with excess ammonia solution, one formed a white precipitate with dil. $NaCl$ solution and one formed a black precipitate with H_2S. The salt could be

(A) $AgNO_3$

(B) $Pb(NO_3)_2$

(C) $Hg(NO_3)_2$

(D) $MnSO_4$

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Q.17

Prussian blue is formed when

(A) ferrous sulphate reacts with $FeCl_3$

(B) ferric sulphate reacts with $K_4Fe(CN)_6$

(C) ferrous ammonium sulphate reacts with $FeCl_3$

(D) ammonium sulphate reacts with NO_2

Q.18

The presence of magnesium is confirmed in the qualitative analysis by the formation of a white crystalline precipitate of

(A) $Mg\,(HCO_3)_2$

(B) $MgNH_4PO_4$

(C) $MgNH_4\,(HCO_3)_3$

(D) $MgCO_3$

Q.19

The sodium extract prepared from sulphanilic acid, contains SCN^-. It gives blood red colouration with

(A) a mixture of $Na_2\ S$ and CS_2

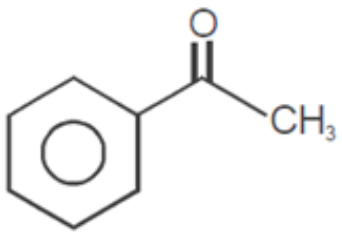
(B) $FeCl_3$

(C) $FeSO_4$

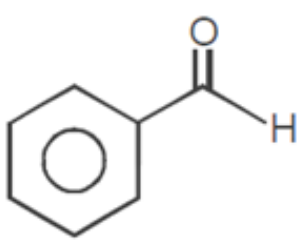
(D) Na_2SO_3

Q.20

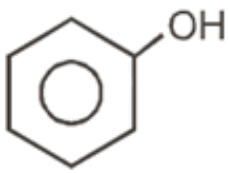
Which of the following is true for the given compounds ?



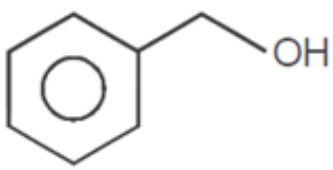
(I)



(II)



(III)



(IV)

(A) (I) and (II) do not give fehling solution test

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(B) (III) Reacts with NaOH but (IV) does not

(C) (I) and (III) give neutral FeCI test but (II) and (IV) do not

(D) All

Answers & Solutions

Q.1 Answer:

I_2

Solution:

Potassium iodide reacts with sulphuric acid to produce potassium sulphate, iodine, hydrogen sulfide and water. Potassium iodide - solid. Sulfuric acid concentrated solution. The reaction takes place in a boiling solution.

$$8KI + 5H_2SO_4 \xrightarrow{\Delta} 4\,K_2SO_4 + 4I_2 + H_2\,S + 4H_2O$$

Q.2 Answer:

$(C_2H_5)_3\,BO_3$

Solution:

$$Na_3BO_3 + \underset{\text{conc.}}{H_2SO_4} \longrightarrow 3Na_2SO_4 + 2H_3BO_3$$

$$H_3BO_3 + 3C_2H_5OH \longrightarrow (C_2H_5)_3\,BO_3 \text{ or } (C_2H_5O)_3\ B$$

Q.3 Answer:

NO_2^-

Solution:

A solution of a salt in dilute sulphuric acid imparts deep blue colour with starch iodine solution it confirms $NO_2{}^-$

Q.4 Answer:

SO_3^{2-} and SO_2

Solution:

Colourlers gas with issitating smell: SO_2 **or** $H_2\,S$.

$$SO_2 + K_2Cr_2O_7 + H_2SO_4 \rightarrow SO_4^{2-} + \underset{\text{Geen}}{Cr^{+3}}$$

$$[x] + H_2SO_4 \rightarrow [y]$$

$$\underset{[x]}{SO_3^2} + \underset{\text{dil.}}{H_2SO_4} \longrightarrow SO_4^{2-} + H_2O + \underset{[y]}{SO_2}\uparrow$$

Q.5 Answer:

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Bromine

Solution:

KBr, on reaction with conc. H_2SO_4**, gives reddish-brown bromine gas.**

$KBr + H_2SO_4 \longrightarrow KHSO_4 + HBr, 2HBr + H_2SO_4 \longrightarrow Br_2 + 2H_2O + SO_2$

Q.6 Answer:

$FeCl_3$

Solution:

Fe^{3+}**, radical gives blood red with** CNS^-

$Fe^{3+} + 3CNS^- \longrightarrow Fe(CNS)_3$

Ferric sulphocyanide (Blood red colouration) Fe^{3+} **gives blue colour on treatment with**

$K_4\,[Fe(CN)_6]$

$4Fe^{3+} + 3\,K_4\,[Fe(CN)_6] \longrightarrow$

$12\,K^+ + Fe_4\,[Fe(CN)_6]_3$

Ferricferrocyanide (Prussian blue) Cl^- **radical gives chromyl chloride test.**

Q.7 Answer:

Co^{2+}

Solution:

Co^{+2} **ion is precipitated by** $H_2\ S$ **in presence of** NH_4OH **which is a group reagent of group IV in cationic analysis.**

Q.8 Answer:

Pb^{2+} salt

Solution:

$PbCl_2$ **is insoluble in cold water, soluble in hot water and PbS is black ppt in acidic medium.**

Q.9 Answer:

Mg^{2+}

Solution:

$MgCl_2$ **is soluble in water and does not give flame test. Other give flame test.**

Q.10 Answer:

$K_3\,[Cu(CN)_4]$ more stable and $K_2\,[Cd(CN)_4]$ less stable

Solution:

$K_3\,[Cu(CN)_4]$ **is more stable whereas** $K_2\,[Cd(CN)_4]$ **is less stable.**

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Q.11 Answer:

Mg^{2+} ions will also be precipitated.

Solution:

If Na_2CO_3 is used in place of $(NH_4)_2\,CO_3$.It will precipitate group V radicals as well as magnesium radicals. The reason for this is the high ionization of Na_2CO_3 in water into Na^+ and CO_3^{2-}. Now the higher concentration of CO_3^{2-} is available which exceeds the solubility product of group V radicals as well as that of magnesium radicals.

Q.12 Answer:

Sulphide of all ions

Solution:

As^{3+} and Cd^{2+} are the radicals of group II, whereas Ni^{2+} & Zn^{2+} are the radicals of group IV. The solubility product of group IV radicals is higher as compared to group II. NH_4OH increases the ionisation of $H_2\,S$ by removing H^+ of $H_2\,S$ as unionisable water. $H_2\,S \rightleftharpoons 2H^+ + S^{2-}$

$$H^+ + OH^- \longrightarrow H_2O$$

Thus excess of sulphide ions are present which leads to the precipitation of all the four ions. Note : HCl decreases ionisation of $H_2\,S$ whereas NH_4OH increases the ionisation of $H_2\,S$.

Q.13 Answer:

Fe^{2+} ion also gives blood red colour with SCN^- ion.

Solution:

Only Fe^{3+} ions give blood red colouration with SCN^- ions.

$Fe^{3+} + SCN^- \longrightarrow [Fe(SCN)]^{2+}$

(dark red)

Q.14 Answer:

Zn

Solution:

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$$Zn^{2+} + 2NH_4OH \longrightarrow Zn(OH)_2 + 2NH_4^+$$

White ppt.

$$Zn(OH)_2 + 2NH_4OH \longrightarrow$$

$$(NH_4)_2ZnO_2 + 2H_2O$$

Soluble

$$(NH_4)_2ZnO_2 + H_2S \longrightarrow ZnS + 2NH_4OH$$

White ppt.

Q.15 Answer:

$4.9\ g$

Solution:

The ionic equation for oxidation of Mohr's salt is

$Fe^{2+} \longrightarrow Fe^{3+} + e^-$

Now Eq. of Mohr's salt $= \frac{392}{1} = 392$ **Strength = Normality** $\times$ **Eq. mass**

$= \frac{1}{20} \times 392 = 19.6\ g/lit$

Thus for preparing $250ml$ **of** $N/20$**Mohr 's salt solution, Mohr's salt needed**

$= \frac{19.6}{1000} \times 250 = 4.9\ g$

Q.16 Answer:

$Pb\,(NO_3)_2$

Solution:

$Pb\,(NO_3)_2 + 2NH_4OH \rightarrow$

$Pb(OH)_2 \downarrow + 2NH_4NO_3$

(white ppt) $Pb\,(NO_3)_2 + 2HCl \rightarrow PbCl_2 \downarrow + 2HNO_3$

(white ppt) $Pb\,(NO_3)_2 + H_2\,S \rightarrow PbS \downarrow + 2HNO_3$

(black)

Q.17 Answer:

ferric sulphate reacts with $K_4Fe(CN)_6$

Solution:

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$2Fe_2\left(SO_4\right)_3+3\,K_4\left[(Fe(CN)_6\right]\rightarrow$
$Fe_4\left[Fe(CN)_6\right]_3+6\,K_2SO_4$

prussian blue

Q.18 Answer:

$MgNH_4PO_4$

Solution:

$MgCl_2+Na_2HPO_4+NH_4OH\rightarrow$
$Mg\left(NH_4\right)PO_4(\mathit{whiteppt})+2NaCl+H_2O$

Q.19 Answer:

$FeCl_3$

Solution:

$SCN^-+Fe^{3+}\longrightarrow Fe(SCN)_3(\mathit{bloodredppt})$

Q.20 Answer:

All

Solution:

No Solution available

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